

Physics-informed deep learning for predicting optical properties of Nb-Ta anodic oxide films from spectroscopic ellipsometry

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ABSTRACT

The prediction of optical properties for anodic oxide films from spectroscopic ellipsometry data constitutes a typical inverse problem. Traditional iterative fitting methods suffer from high computational cost and strong dependence on the model structure. This study develops a physics-informed deep learning framework that addresses this challenge through two complementary modeling scenarios. At its core is a Fresnel-constrained neural network, which predicts optical constants (n , k) and ellipsometric parameters [$\tan(\Psi)$, $\cos(\Delta)$] using the two inputs of base metal composition and anodizing voltage. A hybrid loss function involving the Fresnel equations is employed to guarantee physical consistency with optical reflection rules, thus avoiding non-physical solutions. In addition, two complementary data-optimization strategies were introduced: wavelength selection and model-based data augmentation. The wavelength-selection strategy compresses the spectral input by retaining the most informative 200–350 nm region, while Gaussian-noise-enhanced synthetic samples are generated to expand the training dataset and reduce the risk of surrogate-model bias propagation. The integrated model exhibits good prediction accuracy, with test-set RMSE values of 0.090 for n and 0.038 for k . Notably, the model maintains its training and prediction efficiency even as the volume of data increases. These results show that embedding fundamental optical laws into a deep learning structure yields a robust and efficient framework. Together with targeted data-optimization strategies, this framework offers a promising avenue for the high-throughput inverse design and characterization of complex functional oxide systems.

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I. INTRODUCTION

Ellipsometric spectroscopy is a classic and irreplaceable tool for measuring thin-film optical parameters.¹ However, traditional ellipsometric fitting struggles with compositionally graded thin films, failing to guarantee physically consistent and unique solutions in inverse analysis.^{2,3} First, the traditional fitting relies on predefined models and fixed parameters, and it cannot adequately handle complex multilayers or non-ideal surfaces, nor capture microstructural changes in emerging materials like nanofilms and heterojunctions.⁴ Furthermore, this method involves heavy computation and long runtime. Low analysis efficiency

becomes a major limitation, especially for large experimental datasets.

In recent years, the development of deep learning technologies, particularly the rise of advanced algorithms like Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN),⁵ has provided new perspectives and tools for ellipsometric data processing.^{2,6,7} Increasing research has emerged in this area, including traditional machine learning for film thickness prediction,^{8–10} deep learning for high-precision inversion of n and k ,^{11,12} and notable advances in the generalization ability of physics-constrained models.¹³ Our group prepared anodic oxides

on binary combinatorial libraries of Nb–Ta and Nb–Ti thin films using co-sputtering (Nb concentration range of 10–90 at. %, anodization voltage range of 1–10 V).¹⁴ A deep learning approach was developed to predict n and k of the oxide films directly from ellipsometry data, bypassing the traditional ellipsometric modeling and fitting process.¹⁵ The CNN method performed excellently, achieving RMSE of 0.094 for n and 0.037 for k . We also developed a model to predict Ψ , Δ , n , k at unmeasured composition–voltage combinations.

Existing data-driven models rely heavily on pure data fitting without embedding physical constraints, often producing non-physical results such as negative refractive indices or discontinuous spectra.¹⁶ Especially under unknown compositions or extreme process conditions, the physical plausibility of prediction results is difficult to guarantee. The SUNDIAL method proposed by Liu *et al.*,⁹ which is a closed-loop neural network with an inverse module and a forward module trained as a surrogate for the Fresnel equations. The forward module verifies and constrains the optical constants predicted by the inverse module, enabling high-precision and unique inversion. SENet incorporates Kramers–Kronig relations as constraints to improve physical plausibility and generalization robustness.¹⁷

Concurrently, with the growing demand for automated, rapid, and high-throughput optical characterization of thin-film materials, research on how to further improve the speed and accuracy of ellipsometric data fitting through deep learning model optimization or data processing strategies has become an active area.¹⁸ For instance, Zhu *et al.*¹⁹ further proposed a lightweight 8-bit version of the SUNDIAL model based on the original 32-bit version. By employing a parameter quantization method, DoReFa-Net,²⁰ for training CNN with low-bit-width weights and activations, quantized and compressed the neural-network modules to enhance operational speed. This method successfully compressed the model size from 188.5 to 46.6 MB, significantly reducing computational resource requirements, while processing time decreased from 4.96 to 3.59 s.¹⁹ Besides model improvements, certain data processing strategies are equally crucial for enhancing model performance. Each set of ellipsometric measurement data covers a broad wavelength range (typically from UV–vis to IR). Using the full spectrum without selection increases computational load and may introduce noise from feature-weak bands, degrading prediction accuracy.²¹ Arunachalam *et al.* found that for grade 1 (3 nm thickness accuracy) and grade 2 (1 nm thickness accuracy) classification tasks, models trained using only the first 250 and 300 wavelength data points, respectively, achieved optimal accuracy, comparable to or even better than using uncompressed data. Compression rates of 62.1% and 54.6% cut redundancy and computation. Therefore, selecting proper wavelength intervals as model input is a key to optimize performance.²²

Furthermore, we need to consider the complexity of ellipsometric data fitting, the high cost of sample preparation and the large data demand of deep learning. It is, therefore, difficult to acquire sufficient high-quality data to prevent overfitting and ensure model generalization. Kwak *et al.*¹¹ solved the issue of insufficient experimental data, generated 1000 simulated points by amplifying layer thickness variations. They also generated 1800 augmented data points from 18 normal samples using a noise injection method.²³

Based on the research background described above, this study further improves the optical parameter prediction performance of an existing CNN model through the following three aspects: (1) We incorporate the Fresnel equations as a soft constraint into the loss function to build a physics-informed deep learning model. This effectively prevents non-physical solutions and enhances the physical plausibility and interpretability of the model outputs.²⁴ (2) Addressing the high redundancy in full-spectrum ellipsometric data, we systematically analyze the impact of different wavelength intervals on model performance. By selecting informative wavelength bands, we obtain effective data compression. It makes training and inference faster, while keeping prediction accuracy almost the same. (3) Even with a high-throughput method for preparing composition-gradient samples, the amount of available data remains insufficient for deep learning. We, therefore, used a pre-trained model for unmeasured points to generate synthetic data across the composition–voltage space, augmenting the training set. This enhances the model's generalization ability and prediction stability. This provides us a more practical way to apply the deep learning method to ellipsometric data analysis and hopefully help this method more widely used in the optical thin-film field.²⁵

II. MODELS AND METHODS

A. Sample preparation and spectroscopic ellipsometry

Details on the preparation of Nb–Ta and Nb–Ti anodic oxide films can be found in. Ref. 15. The prepared oxide films show lateral compositional gradients and longitudinal voltage differences.²⁶ Ellipsometric parameters Ψ and Δ were measured via a variable-angle spectroscopic ellipsometer (SE UV 800), and the scanning wavelength set from 200 to 900 nm. A total of 484 datasets were obtained (335 for Nb–Ta, 149 for Nb–Ti). Ψ and Δ are the two fundamental ellipsometric parameters, representing the amplitude ratio and phase difference between the p - and s -polarized reflected light, respectively. Mathematically, these parameters are defined via the complex reflectance ratio ρ as shown in Eq. (1), where ρ is the ratio of the complex Fresnel reflection coefficients for p - and s -polarized light. The Fresnel coefficients r_p and r_s are determined by the complex refractive indices of the ambient, oxide film, and substrate, as well as the film thickness and angle of incidence. Equation (1), thus, directly links the measured ellipsometric quantities Ψ and Δ to the underlying Fresnel physics. The collected data were fitted using SpectraRay4 software to acquire the thickness d , refractive index $n(\lambda)$, and extinction coefficient $k(\lambda)$ of the oxide layers. The obtained d , $n(\lambda)$, and $k(\lambda)$ serve different roles in the subsequent deep learning framework. For the measured-point prediction models (including the Fresnel model, data-compression model, and comprehensive model), d is not a model output; rather, it is a known parameter derived from SpectraRay4 fitting and is used exclusively as an input to the Fresnel optical computation module (Sec. III A) to compute the physically constrained ellipsometric parameters $\tan(\Psi)$ and $\cos(\Delta)$. For the unmeasured-point prediction model, d is predicted as a separate scalar quantity alongside E_g , independent of the wavelength-dependent $n(\lambda)$ and $k(\lambda)$ predictions. This design ensures that the thickness–refractive index correlation

ambiguity—which is a well-known challenge in traditional ellipsometric fitting—does not affect our architecture. Detailed fitting details are described in the literature.^{14,15,27} Ellipsometric measurement and fitting data constitute the dataset used in this work. Please refer to Table S1 in the [supplementary material](#) for the physical definitions of all symbols,

$$\rho = \frac{r_p}{r_s} = \tan(\Psi) \cdot e^{i\Delta}. \quad (1)$$

B. Deep learning methodology

The dataset was randomly divided into a training set (80%) and an independent testing set (20%) using a fixed random seed of 42. During model training, 10% of the training data (i.e., 8% of the total dataset) was further allocated as a validation subset through the `validation_split` function to monitor model convergence and prevent overfitting. Consequently, the effective proportions were 72% for training, 8% for validation, and 20% for testing. The independent testing set was not used during model training or validation and was only used for final performance evaluation. Model performance was evaluated on the independent testing set using RMSE and R^2 metrics. All input and output variables were normalized prior to training.

All models in this work adopt the CNN architecture. All CNN models consist of 1D convolutional (Conv1D), max-pooling (MaxPooling1D), dropout, flatten, and fully connected (dense) layers. To reduce the dependence on manually selected network configurations, the CNN architectures were optimized using the Optuna hyperparameter optimization framework. The search space included the number of convolutional layers, the number of filters in each layer, convolutional kernel sizes, activation functions, pooling operations, dropout rate, and the number of dense units. In addition, several training-related hyperparameters were optimized simultaneously, including the optimizer type, learning rate, batch size, validation split ratio, L2 regularization coefficient, and the parameters associated with early stopping and learning-rate scheduling. The final network structures used in this study correspond to the hyperparameter combinations that yielded the lowest validation loss during the optimization process. [Table I](#) presents all models

TABLE I. Overview of all adopted models.

Model	Description
Fresnel model	Making the output (n , k) and (Ψ , Δ) satisfy the Fresnel optical laws
Data-compression model	Selecting the most information-rich spectral band
Unmeasured-points model	Predicting (n , k) and (Ψ , Δ) at the unmeasured points of the samples
Data augmentation model	Combining the data predicted by the unmeasured-points model with the test data
Comprehensive model	Combination of data augmentation and data compression

adopted in this work. Because spectral data exhibit strong local correlations between neighboring wavelength points, one-dimensional (1D) convolutional layers are ideal for extracting local spectral features. Figure S1 and Table S2 in the [supplementary material](#) show the network architecture and input–output details of each model as well as the selection of model activation functions and kernel sizes.

The Fresnel module was implemented entirely using TensorFlow tensor operations. Therefore, all Fresnel calculations remained fully differentiable and were compatible with TensorFlow automatic differentiation. Gradients of the physics-based loss were automatically propagated through the Fresnel module during backpropagation.

III. RESULTS AND DISCUSSION

A. Implementing the Fresnel-constrained model

As shown in [Fig. 1](#), the Fresnel model adopts (c_{Nb} , V) as input parameters and outputs n , k , $\tan(\Psi)$, and $\cos(\Delta)$. The relationship conforms to the following equation:

$$f(c, V) = (\tan(\Psi)_{\text{pred}}, \cos(\Delta)_{\text{pred}}, n_{\text{pred}}, k_{\text{pred}}). \quad (2)$$

The parameters obtained from the previous iteration (n^{t-1} , k^{t-1}) are input into the Fresnel Optical Computation Module (please see Part 3 in the [supplementary material](#) for the details) to compute the physically constrained optical parameters $\tan(\Psi)_{\text{Fresnel}}$ and $\cos(\Delta)_{\text{Fresnel}}$.^{28–30} n^{t-1} , k^{t-1} , $\tan(\Psi)^{t-1}$, and $\cos(\Delta)^{t-1}$, together with n , k , $\tan(\Psi)$, and $\cos(\Delta)$, form the L_{MSE} in the loss function, while $\tan(\Psi)^{t-1}$ and $\cos(\Delta)^{t-1}$ paired with $\tan(\Psi)_{\text{Fresnel}}$ and $\cos(\Delta)_{\text{Fresnel}}$ constitute the L_{Fresnel} of the loss function. The total loss function L_{total} is defined as the weighted sum of a mean squared error term L_{MSE} and a physics constraint term L_{Fresnel} , as followed in the following equation:

$$\text{LOSS} = L_{\text{MSE}} + \alpha L_{\text{Fresnel}}. \quad (3)$$

Here, α is the weight coefficient. Backpropagation is applied to compute gradients and update the CNN parameters. A detailed Fresnel gradient propagation is provided in Part 4 in the [supplementary material](#). The logical procedure of the Fresnel model is detailed in Part 5 in the [supplementary material](#).

B. Enhancing physical consistency with Fresnel constraints

While purely data-driven models can effectively map composition and voltage to optical properties, they may produce physically unreasonable results, such as negative extinction coefficients or discontinuous spectra—especially when training data are sparse. Embedding the Fresnel equations into the training process enables the model to obey the laws of reflection while minimizing error, thereby making predictions more reliable.³¹

In the whole model architecture, the key issue is how to balance the data-driven loss and the physics-based loss. To handle this, we introduced a parameter α [see Eq. (2)]. We then tested the model with α values from 0.5 down to 0.025. No obvious instability, oscillatory convergence, or divergence between the data loss

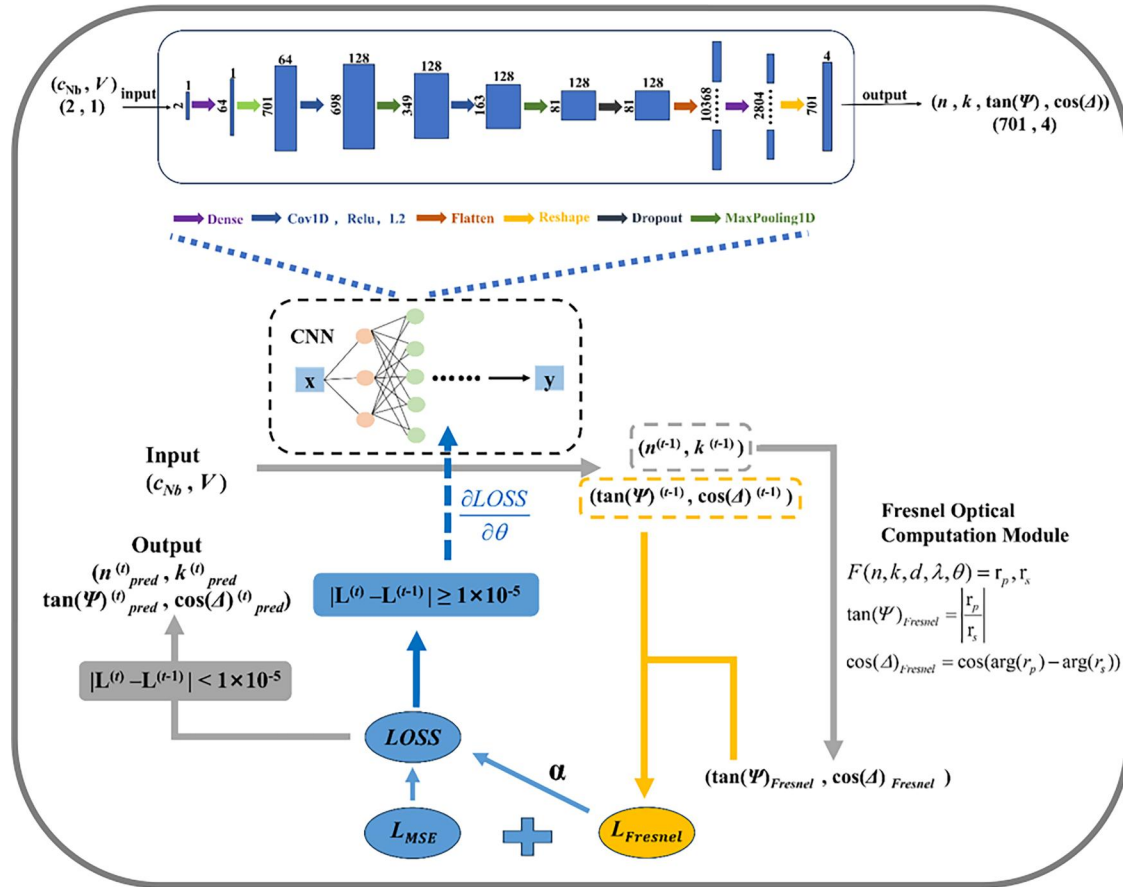


FIG. 1. The architecture of the Fresnel model. The architecture integrates a convolutional neural network (CNN) predictor and a Fresnel optical computation module, enabling the joint optimization of optical constants and ellipsometric parameters through a hybrid loss function that combines data-driven mean squared error (MSE) and physics-based Fresnel constraints.

and physics loss terms were observed. Given the stable convergence observed with the fixed weighting factor, dynamic weighting strategies—including adaptive loss balancing and learning-rate annealing—were not implemented in the current study. Instead, we adopted the fixed $\alpha = 0.05$ configuration, which yielded satisfactory predictive performance and efficient training. We acknowledge that adaptive weighting may further improve robustness and plan to explore it when extending the present framework to more complex oxide systems.

From Fig. 2(a), we can see that the RMSE and R^2 for $\tan(\Psi)$ and $\cos(\Delta)$ change clearly with α . When α is large, like 0.5, the physics constraint dominates and leads to underfitting; and when α is too small, like 0.025, the constraint becomes negligible.

After α is set to the best value, we compared the model with the physics constraint to the original data-driven model. The results are in Fig. 2(b). In terms of RMSE, R^2 , prediction time, and training time, the physics-constrained model performs better.

In terms of prediction accuracy [see Fig. 2(b)], the Fresnel model cuts the RMSE for $\tan(\Psi)$ from 0.026 to 0.022, a drop of 15.4%. For $\cos(\Delta)$, the RMSE goes from 0.063 to 0.057, a drop of

9.5%. At the same time, the R^2 for $\tan(\Psi)$ rises from 0.919 to 0.946 and for $\cos(\Delta)$ from 0.966 to 0.973. In terms of training efficiency, the Fresnel model cuts training time by half compared to the original model. This is because the Fresnel equations guide the predictions toward physically reasonable regions early on. So the model does not spend much time fixing results that break the physics. This makes it converge faster. The prediction times for the two models are close, with a difference of only 1 ms.

We examined how the Fresnel constraint influences model behavior by analyzing prediction errors for n , k , $\tan(\Psi)$, and $\cos(\Delta)$ at different wavelengths as shown in Fig. 3. From Fig. 3(a), we see that the prediction error for n stays around zero from 200 to 900 nm. In the short wavelength range (200–350 nm), the standard deviation is a bit higher. As the wavelength gets larger, the error drops quickly. This shows that the physics constraint helps reduce uncertainty in the inversion process. From Fig. 3(b), the error for k is smaller, and the standard deviation is also narrower. The figure shows a key role of the Fresnel constraint. It stops the model from using non-physical absorption terms to fix fitting

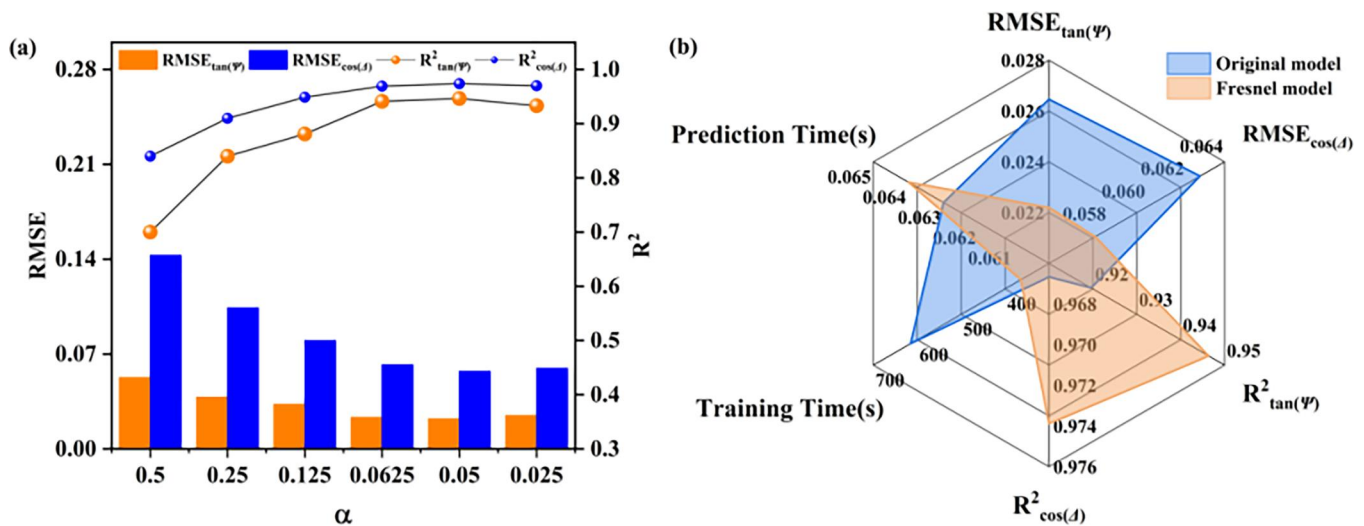


FIG. 2. (a) Comparison of model performance under different weights; (b) comparison of performance between the original model and the physically constrained model.

errors. This is especially true for low-absorption oxide systems like tantalum oxide.³² From Fig. 3(c), the error for $\tan(\Psi)$ is very small and stays stable. This reflects that the constraint works on the relationship between the reflection amplitudes. In Fig. 3(d), the uncertainty bands of $\cos(\Delta)$ are slightly wider in several wavelength regions, particularly below 300 nm and toward the long-wavelength end of the spectrum. This behavior may be related to the increased sensitivity of phase-dependent ellipsometric parameters near Brewster-angle-related and phase-sensitive conditions, where small variations in the optical constants can produce noticeable changes in the phase response. Despite this effect, the mean error remains close to zero and the uncertainty bands remain narrow throughout the investigated wavelength range, confirming the good predictive performance of the model for $\cos(\Delta)$.³³ The fact that the model captures this physical feature actually proves it works well. To further evaluate model generalization, the training and validation loss curves are shown in Fig. S2 in [supplementary material](#) Part 4. Both curves decrease synchronously and converge smoothly, indicating stable optimization and the absence of severe overfitting. To further evaluate the prediction quality at the spectral level, representative comparisons between measured and predicted optical responses were performed for two independent samples with composition–voltage combinations of 44%–9 V and 33%–8 V. The corresponding results are presented in Fig. S3 in the [supplementary material](#). For both samples, excellent agreement is observed between the measured and predicted spectra of n , k , $\tan(\Psi)$, and $\cos(\Delta)$, confirming the capability of the proposed framework to capture both global spectral characteristics and local wavelength-dependent features. Table S3 in the [supplementary material](#) compares the RMSE and R^2 values of the training, validation, and test sets. As summarized in Table S3 in the [supplementary material](#), the model achieves consistently low RMSE and high R^2 values across the training, validation, and test sets. The close

agreement among the three sets—particularly the small gap between validation and test errors—confirms that the model does not suffer from severe overfitting and generalizes well to unseen data.

C. Optimizing data through wavelength selection

The Fresnel-constrained model already gives physically consistent outputs, but the full-spectrum data (701 wavelengths per sample) are highly redundant. Training on all of them costs more computation and may bring in noise from less informative bands. We, therefore, looked for the most informative wavelength range. As shown in Fig. S4 and Part 6 in the [supplementary material](#), the statistical analysis of Ψ , Δ , and their derivatives indicates that most feature variations lie below 350 nm. As shown in Fig. 4, the $\cos(\Delta)$ peak, which is sensitive to composition and voltage, always appears before 350 nm, about 40–110 nm below the mean bandgap (393.8 nm).²⁷ Hence, we focused on the 200–350 nm band.

To validate this choice, we took the first N wavelength points ($N = 50, 100, \dots, 300$) and retrained the measurement-point model for each N [Fig. 5(a)]. Figure 5(b) shows that $N = 150$ gives the best trade-off: RMSE for n drops from 0.094 (full spectrum) to 0.092, and RMSE for k stays 0.037; R^2 for n improves from 0.871 to 0.875. Using only the first 150 points cuts the data volume by 78.6%. On our server (AMD EPYC 7453 + RTX 4090), training time falls from 8.26 to 6.46 s (21.8% faster), and prediction time from 0.33 to 0.24 s (27.3% faster). Accuracy does not degrade; it even improves slightly. Therefore, for the Nb–Ta and Nb–Ti systems, the 200–350 nm band contains the most characteristic ellipsometric information. This selective compression provides a practical way to balance data size, computational load, and prediction performance.³⁴

Although the thickness–refractive index correlation ambiguity is not relevant to our architecture as discussed in Sec. II A, we

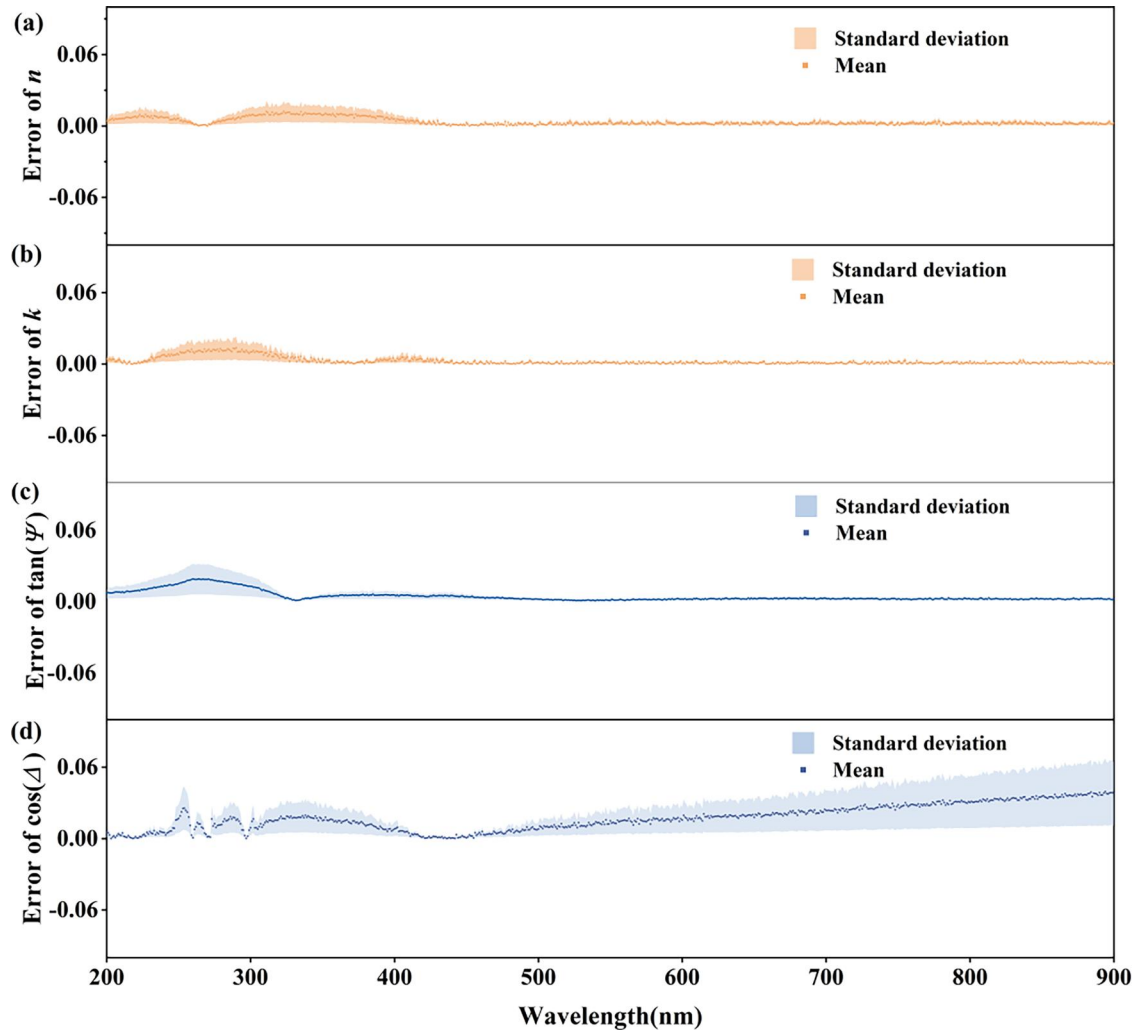


FIG. 3. Mean error and standard deviation between predicted and measured Data: (a) n ; (b) k ; (c) $\tan(\Psi)$; (d) $\cos(\Delta)$.

further verify that wavelength truncation does not degrade prediction accuracy by comparing the full-spectrum and 200–350 nm models. The RMSE of n slightly decreases from full-spectrum model 0.094 to 200–350 nm model 0.092, while the RMSE of k remains 0.037. This result indicates that the 200–350 nm range retains the dominant spectral information required for the present prediction task. The wavelength-selection strategy should be viewed as a data-compression approach for CNN prediction rather than a replacement of conventional full-spectrum ellipsometric fitting.

D. Expanding training data via model-based augmentation

Deep learning models need large, high-quality datasets to avoid overfitting and generalize well.^{35,36} Our composition-gradient samples already reduce preparation work, but the amount

of measured data (484 sets) are still modest for a CNN. Moreover, ellipsometric fitting is expert-dependent and time-consuming. We, therefore, generated synthetic data using two pre-trained unmeasured-point models [one for n and k ; the other for Ψ , Δ ; architectures in Fig. S1(c) in the [supplementary material](#)]. Both models share the same fixed random seed for splitting, so the test set remains untouched and evaluation is consistent.

The two unmeasured-point models perform well:¹⁴ RMSE values for Ψ , Δ , n , and k are 0.919, 16.434, 0.087, and 0.037, respectively, with R^2 of 0.963, 0.921, 0.888, and 0.991. These numbers show that they capture the trends reliably. We then used them to predict Ψ , Δ , n , and k across the whole parameter space c_{Nb} from 1% to 99% in 1% intervals and V from 1 to 10 V in 1 V intervals. This produced 1980 synthetic datasets. Merged with the original 484 measured sets (the test set kept unchanged), we obtained an augmented training set of 2464 samples.

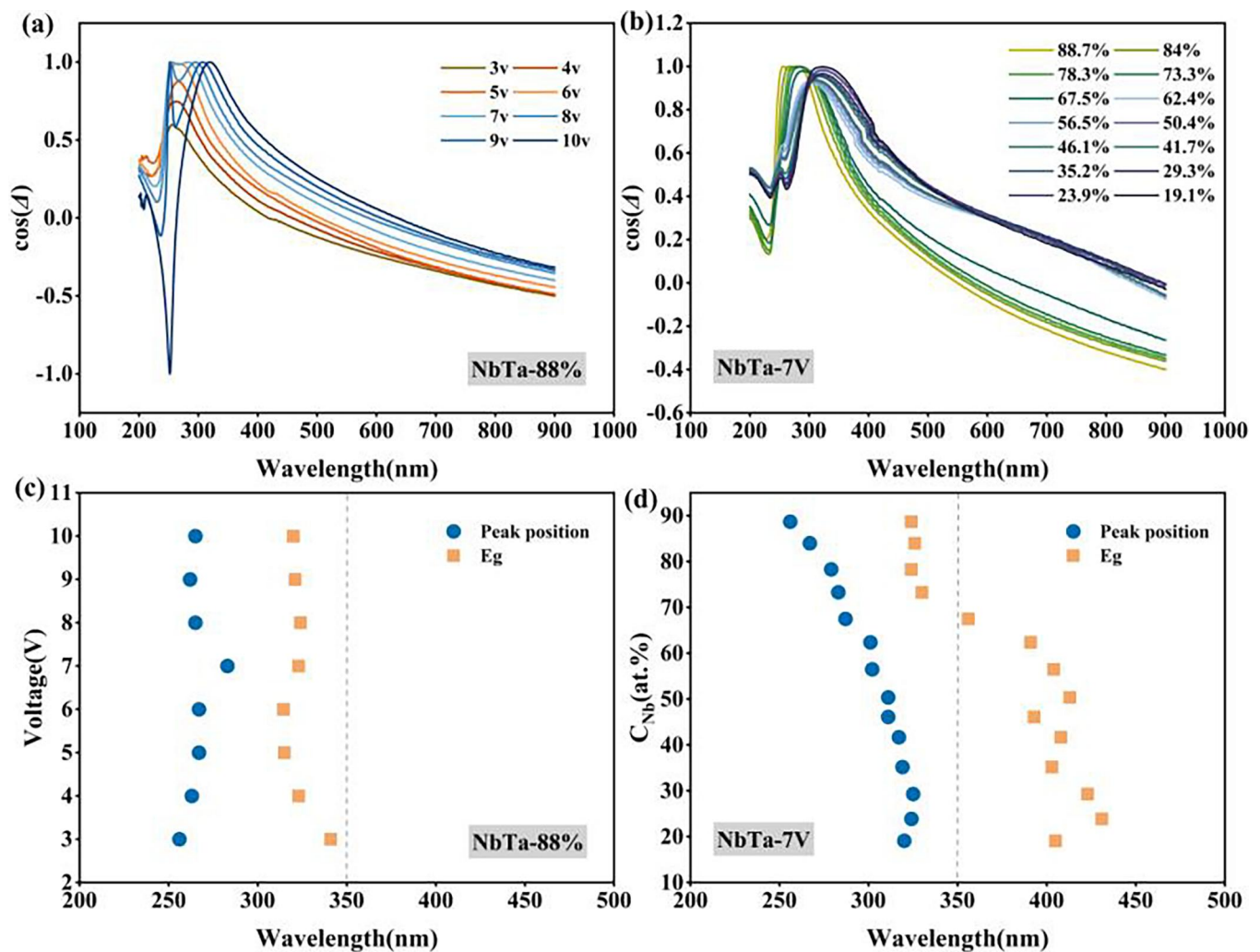


FIG. 4. Correlation between $\cos(\Delta)$ peak features and bandgap energy. $\cos(\Delta)$ curves at varying voltages ($C_{Nb} = 88\%$) (a) and concentrations ($V = 7$ V) (b); $\cos(\Delta)$ peak position vs E_g under varying voltages (c); and concentrations (d).

To reduce the influence of potential prediction bias introduced by the unmeasured-point model, Gaussian noise was added to the generated samples before they were merged with the experimental dataset. Specifically, zero-mean Gaussian noise with a standard deviation equal to 1% of the corresponding variable magnitude was applied to the generated Ψ , Δ , n , and k values. The experimentally measured samples were kept unchanged, and only the generated samples were perturbed. The purpose of this modification is to introduce a reasonable level of uncertainty into the generated data and reduce the possibility that the final model learns the specific prediction errors of the unmeasured-point model. As a result, the CNN is encouraged to learn more general relationships between the input parameters and optical responses rather than reproducing the outputs of the surrogate model. After introducing noise into the synthetic samples, the data

augmentation model achieved RMSE values of 0.100 for n and 0.037 for k , with R^2 values of 0.850 and 0.990, respectively. Hyperparameters were reoptimized with Optuna, giving the architecture in Fig. S1(d) in [supplementary material Part 2](#).

E. Combining wavelength selection and data augmentation

The analysis in Secs. III C and III D highlights two complementary strategies. Wavelength selection maintains accuracy while cutting input dimensions and computational cost; data augmentation expands the training set and boosts generalization ability. Here, we explore whether their combination can further improve model performance.

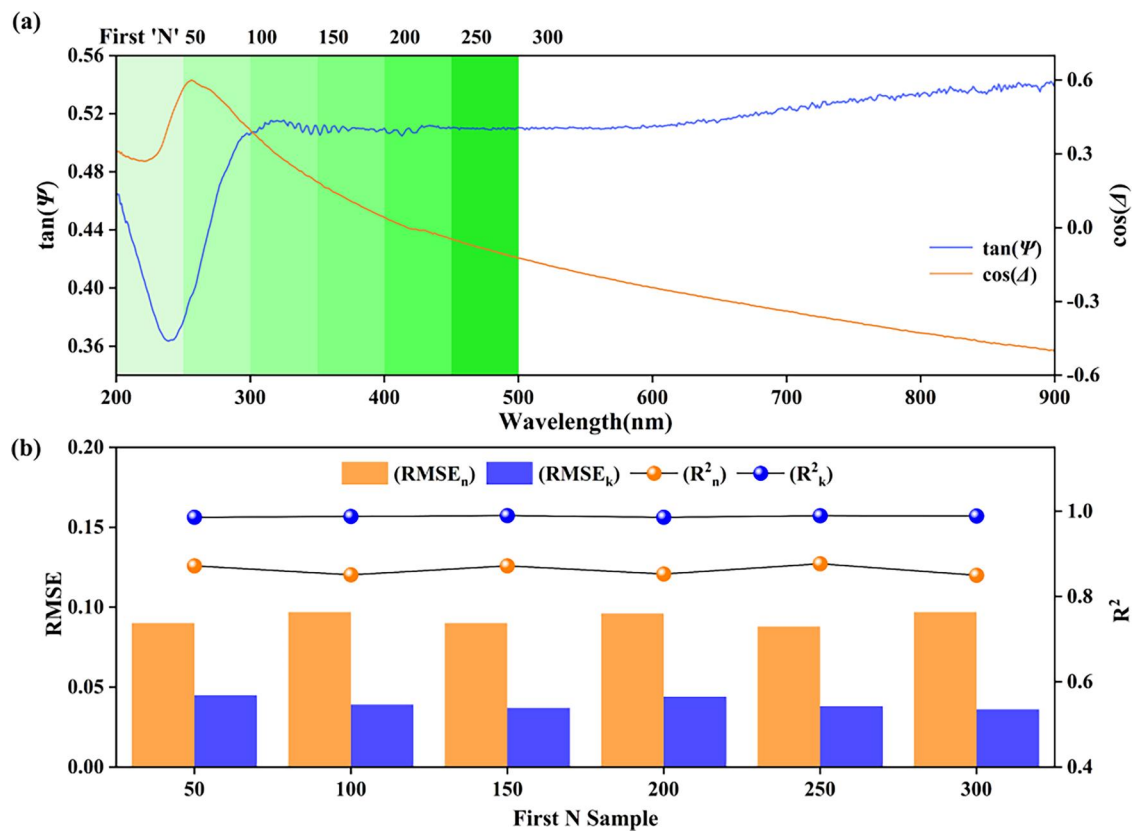


FIG. 5. Comparison of wavelength selection effects: (a) Schematic of the first N data points for Ψ and Δ ; (b) Network architecture of the measurement-point prediction model for $N = 150$.

In the subsequent work, a total of 4444 sample datasets are constructed by combining raw measured data, outputs from the unmeasured-point model, and noise-injected data, and we extract the first 150 wavelength points to form the final comprehensive training dataset. Based on this dataset, we optimize the model hyperparameters and finally determine the structure of the comprehensive model, as shown in Fig. S1(e) in [supplementary material](#) Part 2.

To evaluate the performance of different models, we compare the RMSE and R^2 of the original model, data-compression model, data augmentation model, and comprehensive model on the same test set. Detailed results are listed in [Table II](#). The results show

that the data-compression model improves training speed while maintaining prediction accuracy. The comprehensive model, on the other hand, greatly enhances the prediction accuracy of n with almost no loss of computing speed.

These results indicate that by adjusting the degree of data compression and the intensity of data augmentation, we can effectively balance the model's prediction efficiency and accuracy, thus meeting the practical needs of rapid characterization of optical thin films.

To further compare the difference in n and k prediction performance between the comprehensive model and the original model, the average loss curves for n and k predicted by both

TABLE II. Performance comparison between the original model and optimized model.

Model	RMSE- n	RMSE- k	R^2-n	R^2-k	Training time (s)	Prediction time (s)
Original model	0.094	0.037	0.871	0.991	8.26	0.33
Data-compression model	0.092	0.037	0.875	0.991	6.46	0.24
Data augmentation model	0.101	0.037	0.850	0.991	24.05	0.54
Comprehensive model	0.090	0.038	0.884	0.991	14.69	0.23

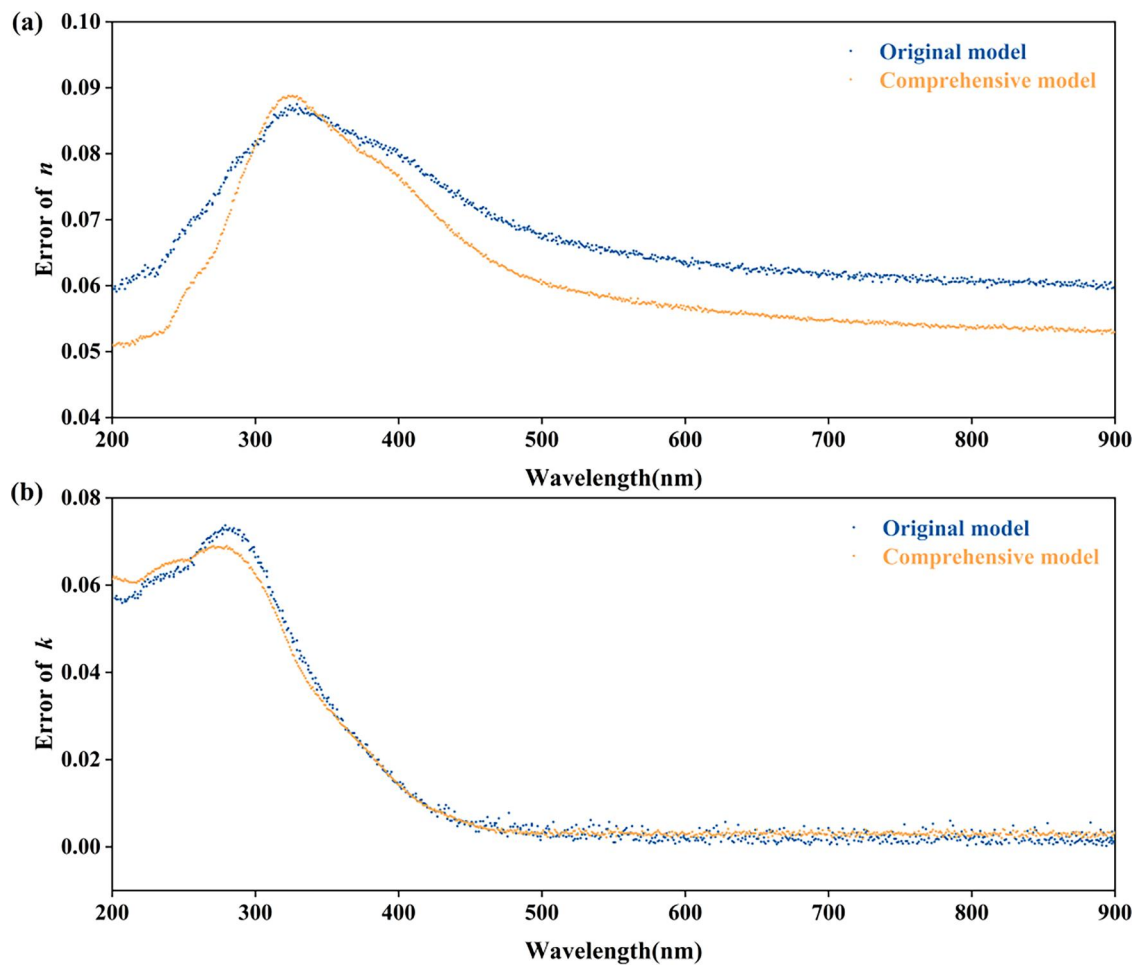


FIG. 6. Average loss for n (a) and k (b) predicted by the comprehensive model vs the original model at each wavelength.

models at each wavelength were plotted, as shown in Fig. 6. It can be seen that the comprehensive model exhibits significantly lower loss in predicting n , while its loss for predicting k is mainly reduced around the loss peak near 300 nm. The reason for this phenomenon might be that the original model's prediction loss for k is already below 0.02 above 400 nm, leaving limited room for further improvement.

IV. CONCLUSIONS

This study developed a physics-informed deep learning architecture. It predicts optical constants (n , k) and ellipsometric parameters (Ψ , Δ) of anodic oxide films from spectroscopic ellipsometry data. Three complementary optimization strategies were used to achieve accurate, efficient, and physically consistent predictions.

First, we incorporated the Fresnel equations into the loss function to constrain the model to follow the laws of reflection. A pure data-driven model was used for comparison. This new model

reduced the RMSE of $\tan(\Psi)$ by 15.4% and the RMSE of $\cos(\Delta)$ by 9.5%. It also cut the training time by nearly half. The second strategy focus on was selecting informative wavelengths. We found that the 200–350 nm range has the most useful information. We, therefore, used these 150 wavelengths for model training. This reduced the data size by 78.6% and sped up training by 38.9%. Despite the large reduction in data size, prediction accuracy did not degrade. In fact, the RMSE for n dropped slightly from 0.101 to 0.090. The third strategy was data augmentation. We used the model to generate 3960 synthetic samples for unmeasured points, bringing the total number of training samples to 4444. As a result, the RMSE for n was further reduced to 0.090, and the model performed better on unseen data.

These three strategies tackle different parts of the modeling problem: physical consistency, computational efficiency, and data scarcity. They also complement each other naturally. The final model gave an RMSE of 0.090 for n and 0.038 for k . The R^2 values were 0.850 for n and 0.991 for k . The training time was 14.69 s.

Notably, combining wavelength selection and data augmentation offsets the extra computational cost of augmented training. This shows that data-level optimizations can be used together without losing efficiency.

Beyond improving numerical results, this work offers two insights. Embedding optical physics into deep learning enhances prediction interpretability, and selecting the wavelength range based on the material's bandgap improves both accuracy and efficiency. Future work could use uncertainty-weighted data augmentation to limit error propagation from unmeasured-point models. Overall, this deep learning approach provides a practical route for the high-throughput optical characterization of complex oxides.

SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for the following: (1) Detailed network architectures of all CNN models (Figure S1); (2) training and validation loss curves of the Fresnel-constrained model (Figure S2); (3) representative comparisons between predicted and measured optical responses for two independent samples (Figure S3); (4) wavelength-dependent statistical analysis of ellipsometric parameters and their derivatives (Figure S4); (5) a complete derivation of the Fresnel optical computation module, including gradient propagation formulas for the physics-informed loss function; (6) definitions of all optical and neural-network symbols (Table S1); (8) input-output configurations, activation functions, kernel sizes, and wavelength-selection criteria for all models (Table S2); and (8) performance metrics of the Fresnel model on training, validation, and test sets (Table S3).

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

X.F. and X.H. contributed equally to this paper.

Xinyi Fan: Data curation (equal); Formal analysis (equal); Writing – original draft (equal). **Xiaocha He:** Data curation (equal); Formal analysis (equal); Investigation (equal); Writing – original draft (equal). **Andrei Ionut Mardare:** Resources (lead). **Li Zhang:** Data curation (supporting); Formal analysis (supporting). **Shuhao Chen:** Data curation (supporting); Formal analysis (supporting); Investigation (supporting). **Juan Zuo:** Conceptualization (lead); Funding acquisition (lead); Methodology (lead); Supervision (lead); Writing – review & editing (lead).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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